

# RECITATION 1

## PROBABILITY REVIEW, DECISION TREES

10-701: INTRODUCTION TO MACHINE LEARNING

September 4, 2026

## 1 Probability Review

### 1.1 Joints and Marginals

from 10-301/601 S26

Let  $A$  take values  $a_1, a_2$  and let  $B$  take values  $b_1, b_2$ , with  $P(A = a_1) = 0.5$  and  $P(B = b_1) = 0.5$ .

1. Suppose  $a_1$  and  $b_2$  are disjoint (mutually exclusive). Fill in the full joint distribution  $P(A, B)$ , and give  $p(a_1, b_1)$  and  $p(a_1 | b_2)$ .

Disjointness forces  $p(a_1, b_2) = 0$ . Since  $p(a_1) = 0.5$ , all of  $a_1$ 's mass sits in the  $b_1$  column, so  $p(a_1, b_1) = 0.5$ . That exhausts  $p(b_1) = 0.5$ , so  $p(a_2, b_1) = 0$  and  $p(a_2, b_2) = 0.5$ . Finally  $p(a_1 | b_2) = 0/0.5 = 0$ .

| $P(A, B)$ | $b_1$ | $b_2$ |     |
|-----------|-------|-------|-----|
| $a_1$     | 0.5   | 0     | 0.5 |
| $a_2$     | 0     | 0.5   | 0.5 |
|           | 0.5   | 0.5   | 1   |

Note that this is not merely “dependent”: a single zero entry pinned the entire table, and  $A$  now determines  $B$  exactly.

2. Suppose instead that  $A$  and  $B$  are independent. Fill in the same table and give the same two quantities.

Every atomic event gets  $p(a_i, b_j) = p(a_i)p(b_j) = 0.5 \times 0.5 = 0.25$ .

| $P(A, B)$ | $b_1$ | $b_2$ |     |
|-----------|-------|-------|-----|
| $a_1$     | 0.25  | 0.25  | 0.5 |
| $a_2$     | 0.25  | 0.25  | 0.5 |
|           | 0.5   | 0.5   | 1   |

So  $p(a_1, b_1) = 0.25$  and  $p(a_1 \mid b_2) = 0.5$ : conditioning on  $B$  tells us nothing about  $A$ .

3. The two tables have identical marginals. What does that tell you about recovering a joint distribution from its marginals?

Marginals do not determine the joint. The first table is maximal dependence and the second is independence, yet  $p_A$  and  $p_B$  are the same in both, so no procedure that sees only the marginals could tell them apart. This is why the factorization rules are stated over the joint rather than over marginals.

## 1.2 Distributions and Densities

1. Let  $X \sim \text{Uniform}(0, \frac{1}{10})$ , with density  $p_X$ . What is  $p_X(0.05)$ ? What is  $P(X = 0.05)$ ? Reconcile the two answers.

The density is  $p_X(x) = 10$  on the support, so  $p_X(0.05) = 10$ . But  $P(X = 0.05) = 0$ , since under a continuous density every atomic event has probability zero. There is no contradiction: a density is a probability *per unit length*, not a probability. It is free to exceed 1, and it is constrained only by  $\int_{a \in U} p(a) da = 1$ . This is the distinction the slides draw between a distribution, where  $p(a) \leq 1$  for every atomic event, and a density, where  $p(a)$  can be arbitrarily large.

2. Is  $p(x) = \frac{1}{2\sqrt{x}}$  on  $(0, 1]$  a valid probability density?

Yes. It satisfies  $p(x) \geq 0$  for all atomic events, and

$$\int_0^1 \frac{1}{2} x^{-1/2} dx = [x^{1/2}]_0^1 = 1.$$

It is unbounded as  $x \rightarrow 0^+$ , which is permitted. Only the integral has to equal 1; the pointwise values are unconstrained above.

3. Let  $X \sim \text{Uniform}(0, 1)$  and let  $Y = X^2$ . Derive the density  $p_Y$  and compare it to the previous part.

Work through the probability of the event  $\{Y \leq y\}$ . For  $y \in (0, 1)$ ,

$$P(Y \leq y) = P(X^2 \leq y) = P(X \leq \sqrt{y}) = \sqrt{y},$$

and since the density is the derivative of probability,

$$p_Y(y) = \frac{d}{dy}P(Y \leq y) = \frac{d}{dy}\sqrt{y} = \frac{1}{2\sqrt{y}}.$$

This is exactly the density from the previous part. Squaring a uniform piles mass up near zero, and the density diverges there even though no individual point carries any probability.

### 1.3 Conditional Distributions and Factorization

Consider the conditional distribution  $P(X, Y \mid Z)$  below, and suppose  $P(Z = 0) = 0.4$ .

| $P(X, Y \mid Z)$<br>$X, Y \downarrow$ | $Z \rightarrow$ | 0   | 1   |
|---------------------------------------|-----------------|-----|-----|
| 00                                    |                 | 0.1 | 0.5 |
| 01                                    |                 | 0.2 | 0.3 |
| 10                                    |                 | 0.6 | 0.2 |
| 11                                    |                 | 0.1 | 0.0 |

1. Compute the joint entries  $p(X = 1, Y = 0, Z = 0)$  and  $p(X = 1, Y = 0, Z = 1)$ . Which factorization did you use?

Factor in the order  $Z$ , then  $(X, Y)$ :  $P(X, Y, Z) = P(X, Y \mid Z) P(Z)$ . Evaluating at the atomic events,

$$p(1, 0, 0) = 0.6 \times 0.4 = 0.24, \quad p(1, 0, 1) = 0.2 \times 0.6 = 0.12.$$

2. Compute  $P(Z = 0 \mid X = 1, Y = 0)$ .

Marginalize, then divide. We have

$$\begin{aligned} p(X = 1, Y = 0) &= p(X = 1, Y = 0, Z = 0) + p(X = 1, Y = 0, Z = 1) \\ &= 0.24 + 0.12 = 0.36. \end{aligned}$$

Thus,

$$P(Z = 0 \mid X = 1, Y = 0) = \frac{P(X = 1, Y = 0, Z = 0)}{P(X = 1, Y = 0)} = \frac{0.24}{0.36} = \frac{2}{3}.$$

This is Bayes' rule read directly off the factorization: we were handed  $P(X, Y \mid Z)$  and inverted it to obtain  $P(Z \mid X, Y)$ .

3. Sum the row  $(X, Y) = (1, 0)$  across the two columns. Why is it not 1, and what *is* guaranteed to sum to 1 in this table?

The row sums to  $0.6 + 0.2 = 0.8$ . There is no reason for it to sum to 1: the two entries are conditioned on different events, so they belong to two different normalized distributions and adding them is meaningless. What is guaranteed is that each *column* sums to 1, since  $P(X, Y \mid Z = z)$  normalizes over  $X, Y$  for each fixed atomic event  $z$ , but not the other way around. Conditional distributions normalize over the argument, never over the conditioning variable.

## 1.4 Expectation

1. **Tail-sum formula.** Let  $X$  be a nonnegative integer-valued random variable. Show that

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} P(X \geq k).$$

credit: 10-406/605 F25

Start from the definition and rewrite  $k = \sum_{j=1}^k 1$ :

$$\mathbb{E}[X] = \sum_{k=0}^{\infty} k p(k) = \sum_{k=0}^{\infty} \left( \sum_{j=1}^k 1 \right) p(k).$$

Swap the order of summation:

$$\mathbb{E}[X] = \sum_{j=1}^{\infty} \sum_{k=j}^{\infty} p(k) = \sum_{j=1}^{\infty} P(X \geq j).$$

2. **A distribution that is neither discrete nor continuous.** Let  $Z \sim \mathcal{N}(0, 1)$  and let  $X = \max(0, Z)$ , a ReLU applied to a Gaussian pre-activation. Compute  $P(X = 0)$ ; can  $X$  have a density in the usual sense? Compute  $\mathbb{E}[X]$ .

We have  $P(X = 0) = P(Z \leq 0) = \frac{1}{2}$ . A single atomic event carries positive mass, so no ordinary density exists: any candidate  $p$  would have to integrate to  $\frac{1}{2}$  over the single point  $\{0\}$ .

Note that  $p_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ . The atomic event at zero doesn't contribute to the expectation, so

$$\mathbb{E}[X] = \int_0^{\infty} z p_Z(z) dz = [-p_Z(z)]_0^{\infty} = p_Z(0) = \frac{1}{\sqrt{2\pi}}.$$

## 1.5 Bayes' Rule

You are training a binary classifier to detect spam emails! We know that only 1 in 1000 emails are actually spam. You train a classifier that outputs positive label with:

- probability 0.99 if the sample is truly positive.
  - probability 0.05 if the sample is truly negative.
1. If the classifier predicts “positive” for a new sample, what is the probability that it is actually positive (True Positive)?

**Given Information:**  $P(\text{Spam}) = 0.001$ ,  $P(\text{Not Spam}) = 0.999$ ,  $P(\text{Positive}|\text{Spam}) = 0.99$ ,  $P(\text{Positive}|\text{Not Spam}) = 0.05$

**Part 1:  $P(\text{Spam} | \text{Positive})$**

Using Bayes' theorem:  $P(\text{Spam}|\text{Positive}) = \frac{P(\text{Positive}|\text{Spam}) \cdot P(\text{Spam})}{P(\text{Positive})}$

First, calculate  $P(\text{Positive})$  using total probability:  $P(\text{Positive}) = 0.99 \times 0.001 + 0.05 \times 0.999 = 0.00099 + 0.04995 = 0.05094$

Now apply Bayes' theorem:  $P(\text{Spam}|\text{Positive}) = \frac{0.99 \times 0.001}{0.05094} = \frac{0.00099}{0.05094} \approx 0.0194$

**Answer:**  $P(\text{Spam}|\text{Positive}) \approx 1.94\%$

2. Why is this probability so small even though the classifier is “99% accurate” on positive cases?

The probability is small due to the **base rate fallacy**. Even though the classifier has 99% accuracy on spam emails, the extremely low prevalence of spam (0.1%) means that most positive predictions are false positives.

**Intuitive explanation:** Out of 1000 emails: 1 is truly spam  $\rightarrow$  correctly identified as positive; 999 are not spam  $\rightarrow 999 \times 0.05 \approx 50$  incorrectly identified as positive.



So we get approximately 51 positive predictions, but only 1 is actually spam:  $\frac{1}{51} \approx 1.96\%$

This demonstrates that high sensitivity (99%) doesn't guarantee high precision when the base rate is very low.

## 2 Decision Trees

### 2.1 Entropy

Recall from lecture that the entropy of a distribution for a discrete random variable  $X$  is defined as

$$H(X) = \sum_c -P(X = c) \log_2 P(X = c)$$

Let us define a random variable  $X \in \{0, K - 1\}$ . Define  $\alpha_k = P(X = k)$ . Show that the uniform distribution maximizes the entropy, that is:

$$\max_P H(X) = P(X = k) = \begin{cases} \frac{1}{K} & \text{if } k \in \{0, K - 1\} \\ 0 & \text{else} \end{cases}$$

**HINT:** You will want to introduce a Lagrange multiplier constraint to enforce that the probability terms sum to 1. This constraint is of the form  $\sum_k \alpha_k = 1$ .

#### Approach-1: Lagrange Multipliers

We start by writing the entropy, which is the function that we want to maximize, along with the Lagrange multiplier and constraint:

$$f(x) = \sum_k -\alpha_k \log_2 \alpha_k + \lambda \left( \sum_k \alpha_k - 1 \right)$$

We take the derivative with respect to a specific  $\alpha_k$ :

$$\begin{aligned} \frac{d}{d\alpha_k} f(x) &= \frac{d}{d\alpha_k} \left[ \sum_k -\alpha_k \log_2 \alpha_k + \lambda \left( \sum_k \alpha_k - 1 \right) \right] \\ &= -\log \alpha_k - 1 + \lambda \end{aligned}$$

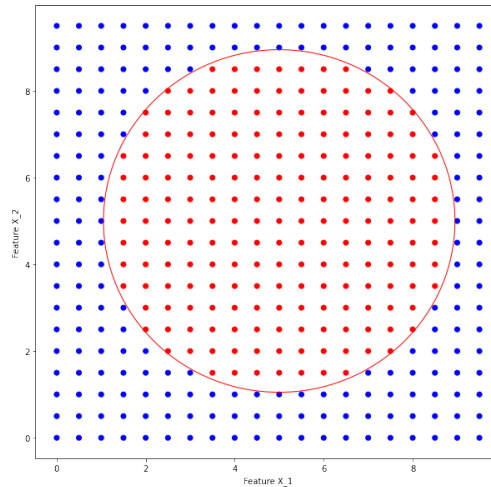
Setting to 0 and solve for  $\alpha_k$ :

$$\begin{aligned} -\log \alpha_k - 1 + \lambda &= 0 \\ -1 + \lambda &= \log \alpha_k \\ \alpha_k^* &= e^{\lambda-1} \end{aligned}$$

Note that the right hand side does not depend on  $k$ , so  $\alpha_k = e^{\lambda-1}, \forall k$ . This means that all  $\alpha_k$  are equal to each other, and with the additional constraint that they must sum to 1, we conclude that  $\alpha_k = \frac{1}{K}, \forall k$ .

## 2.2 Decision Tree Decision Boundaries

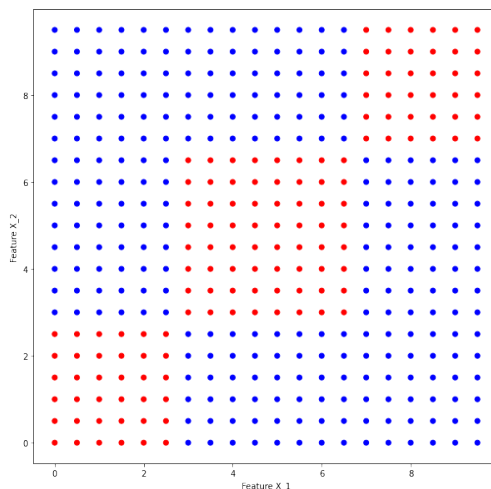
1. Can a learned, finite decision tree using the ID3 algorithm perfectly classify all of the points in the figure below? Can it perfectly learn the decision boundary (portrayed as a circle in red)?



It can classify the points, but cannot perfectly learn the decision boundary.

A decision tree can perfectly classify the points above, but the depth of the tree would be huge. This is because the decision boundary of a decision tree must be axis-aligned. This means that we cannot perfectly learn a decision boundary that is described by a circle.

2. Consider the dataset below with 400 total points consisting of three clusters of red points with 36, 64, and 36 points going from bottom left to top right. The mutual information represents how much we gain, in terms of reducing the entropy, from knowing something about an attribute:  $I(Y; X) = H(Y) - H(Y|X)$ , where  $H(Y|X) = \sum_k p(X = k)H(Y|X = k)$  represents the conditional entropy. Starting with the base predictor, what is the information gain (or mutual information) of the split  $X_1 < 3$ ?

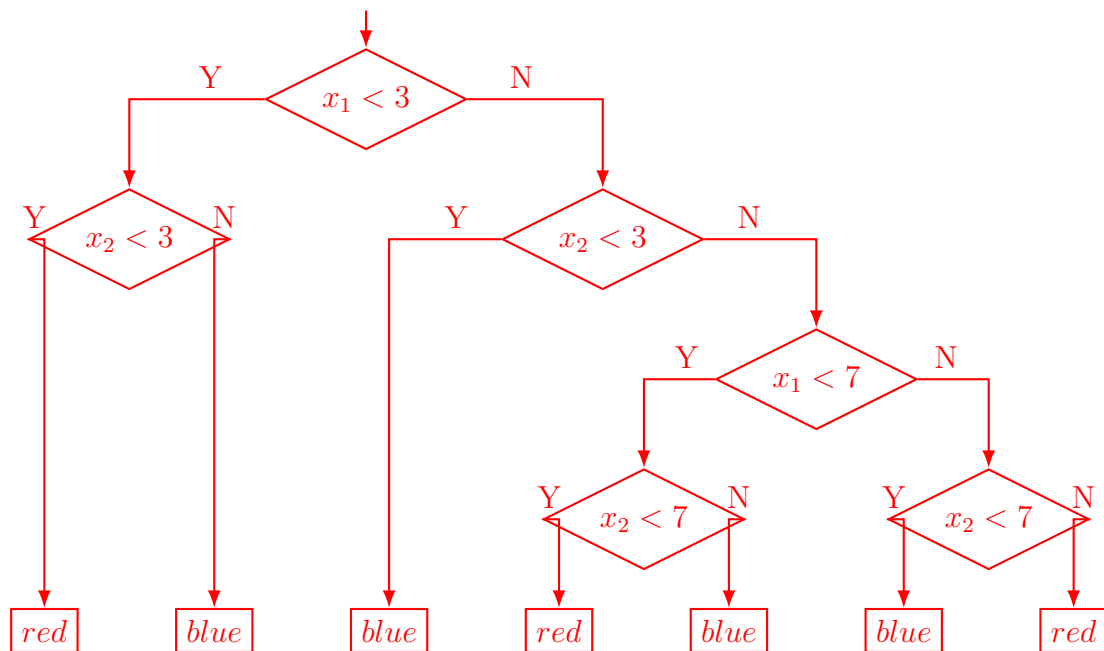


There are 136 red points and  $400 - 136 = 264$  blue points. If we let  $Y$  denote the observed label distribution, then  $H(Y) = -\frac{136}{400} \log_2 \frac{136}{400} - \frac{264}{400} \log_2 \frac{264}{400} \approx .925$ . For points such that  $X_1 < 3$ , there are 36 red points and 84 blue points for a total of 120 points. So  $H(Y|X_1 < 3) = -\frac{36}{120} \log_2 \frac{36}{120} - \frac{84}{120} \log_2 \frac{84}{120} \approx .881$ . For points with  $X_1 \geq 3$ , there are 100 red points and 180 blue points for a total of 280 points. So  $H(Y|X_1 \geq 3) = -\frac{100}{280} \log_2 \frac{100}{280} - \frac{180}{280} \log_2 \frac{180}{280} \approx .94$ . The total conditional entropy is  $H(Y|X_1) = \frac{120}{400}(.881) + \frac{280}{400}(.94) = .9223$ , so the mutual information is  $I(Y; X_1) = H(Y) - H(Y|X_1) = .0027$ .

3. What is the mutual information of the split  $X_2 < 3$  (we are performing this split AFTER the previous split, i.e. on the points with  $X_1 < 3$ ).

There are 120 points in this split and from above we know the entropy is .881. Note that the additional split of  $X_2 < 3$  perfectly classifies the data in this region. Letting  $Y'$  denote the distribution of points with  $X_1 < 3$ , we have that the information gain is  $IG(Y'|X_2) = H(Y') - H(Y'|X_2) = .881 - \frac{36}{120}(0) + \frac{84}{120}(0) = .881$ .

4. Draw out a decision tree that could perfectly classify the points in the figure from Question 2. Why can we exactly learn the decision boundary in this case?



We can exactly learn the decision boundary because it is axis-aligned.

## 2.3 Bonus: KL Divergence

A concept that may be useful for your homework is the KL divergence, a measure of the difference between two probability distributions. For discrete distributions  $p, q$  with support  $\{1, \dots, n\}$ , the KL divergence is:

$$D(p||q) = \sum_{i=1}^n p(i) \log \frac{p(i)}{q(i)}$$

Show then, that  $D(p||q) \geq 0$  for all  $p, q$ .

**Hint:** You may use this inequality without proof:  $x - 1 \geq \log(x)$ .

Proving that  $D(p||q) \geq 0$  is equivalent to proving  $\sum_{i=1}^n p(i) \log \frac{q(i)}{p(i)} \leq 0$ . Using the given inequality:

$$\sum_{i=1}^n p(i) \log \frac{q(i)}{p(i)} \leq \sum_{i=1}^n p(i) \left( \frac{q(i)}{p(i)} - 1 \right) = \sum_{i=1}^n q(i) - p(i)$$

Since  $p(i)$  and  $q(i)$  are normalized, the sum over all events for both is 1. As such:

$$\sum_{i=1}^n p(i) \log \frac{q(i)}{p(i)} \leq \sum_{i=1}^n q(i) - \sum_{i=1}^n p(i) = 0$$